Fast Human Detection for Indoor Mobile Robots Using Depth Images

Motivation

- Mobile robots possess limited computational resources which are shared among various processes required for autonomous operation, so the human detector must have a small computational footprint.
- The detections has to be robust to variation in pose, occlusion and lighting variation.

Related Work

- Naive way of determining candidate regions in an image is to slide a fixed-size window over the entire image, and treat every region of that window size as a candidate.
- Various methods have been proposed to improve speed over the brute-force sliding window search for extracting candidate regions.
- The most widely used descriptor for human detection is the HOG descriptor. Variant of the original HOG descriptor adapted to depth images called Histograms of Oriented Depth (HOD) is being used on depth image.

Algorithm Overview

The Algorithm uses sub-sampling and graph segmentation detect the individual regions, then filters the regions based on a set of heuristic rules and classifies the regions based on their HOD descriptor.

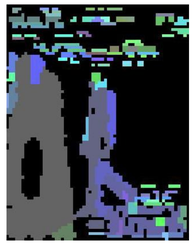
The Algorithm can be grouped into three steps:

1. Depth Image Segmentation
2. Region Filtering and Merging
3.Candidate Classification

Step 1: Image Segmentation

This step accepts a depth image D and returns R, a set of regions corresponding to distinct areas in the depth image.

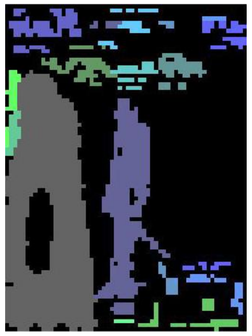
- Depth data from Microsoft Kinect is transformed to Real depth and subsequently to its 3D position using intrinsic parameters and camera field of views.
- The obtained depth image is then divided into small cells, and is sub-sampled to one point to reduce noise and distortion in while obtaining the data.
- The sub-sampled image is then segmented using graphs constructed on Depth and Normal values as edge weights[2].
- Distinct Regions R, are constructed based on the graph segmentation, such that all the points in a region are connected to same segments in both depth and normal graph after graph segmentation.

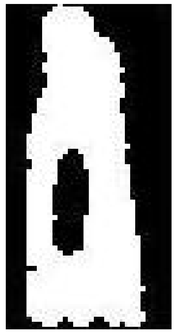


Step 2: Region Filtering and Merging

This step takes a set of regions R as input, discards those a heuristic deems unlikely to be humans, and merges the remaining regions, returning C, a set of candidates likely to correspond to distinct objects.

- For each region, parameters like mean depth, height, width and planarity are calculated and are used to filter regions that are unlikely to be human
- Further the smaller regions are merged with larger regions if they are found to satisfy set a rules.
- The remaining regions are scaled to a fixed size to get a set of candidates C.



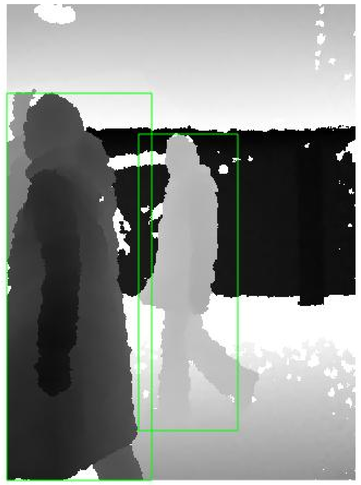


Step 3: Candidate Classification

The last step of our approach takes the set C and returns H, which is the subset of C classified as human.

- We compute the HOD descriptor, where the gradient between any undefined value and some other value is always zero. We use signed gradients since the sign of gradients between metric depths is significant
- we use a Support Vector Machine (SVM) with linear kernel to classify the computed descriptor for a region and decide whether the region contains a human or not.

Results



Why is this fast and Robust ?

- Since we are filtering out the candidates which we are testing, to be humans, it greatly narrows down our search.
- Sub-sampling to reduce noise also means all the operations are performed in a smaller sample of the image, which greatly decreases our computation cost at every step of the algorithm.
- Since the algorithm run on depth data, it is fairly robust to changes in texture and lighting condition.

Reference

- Choi, Byron, et al. "Fast human detection for indoor mobile robots using depth images." Robotics and Automation (ICRA), 2013 IEEE International Conference on. IEEE, 2013.
- Felzenszwalb, Pedro F., and Daniel P. Huttenlocher. "Efficient graph-based image segmentation." International Journal of Computer Vision 59.2 (2004): 167-181.

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